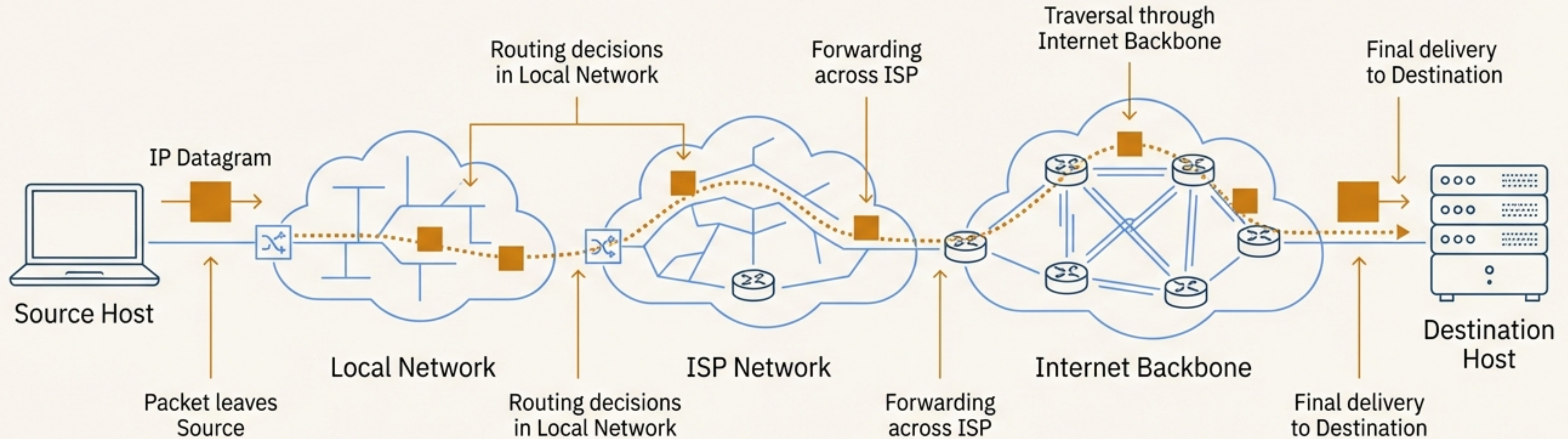


# The Journey of a Packet

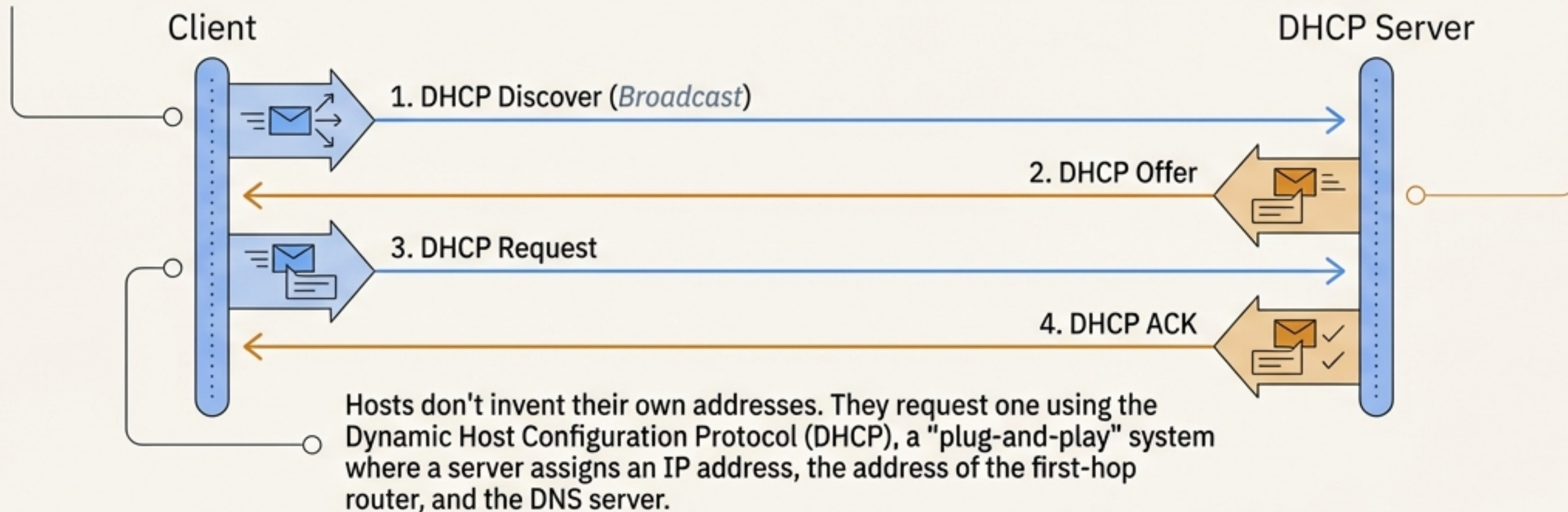
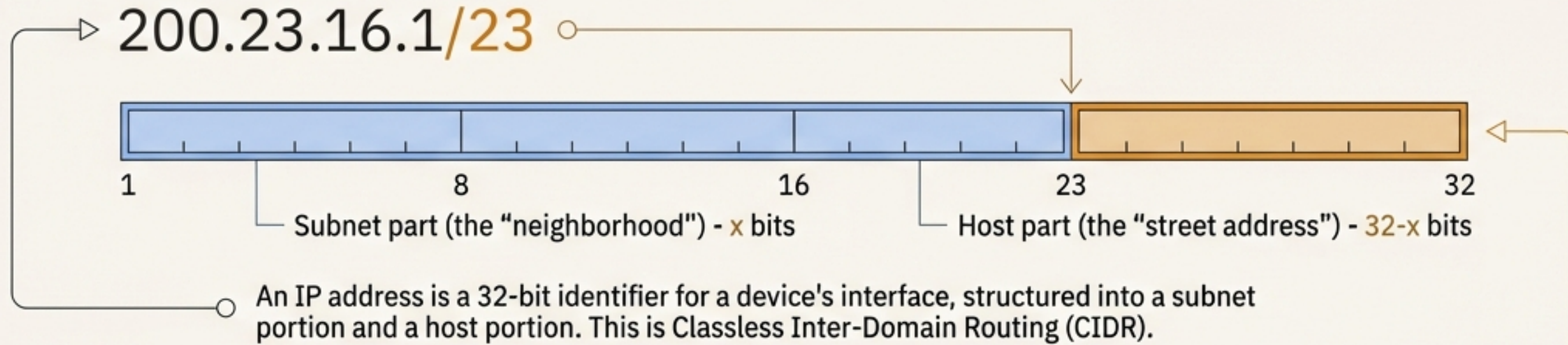
How data travels across the internet's data plane.



Every time you browse a website or send an email, you dispatch data packets on a remarkable journey. We will follow one such packet, an IP datagram, to understand the fundamental mechanics of the network's data plane—the infrastructure that moves your data.

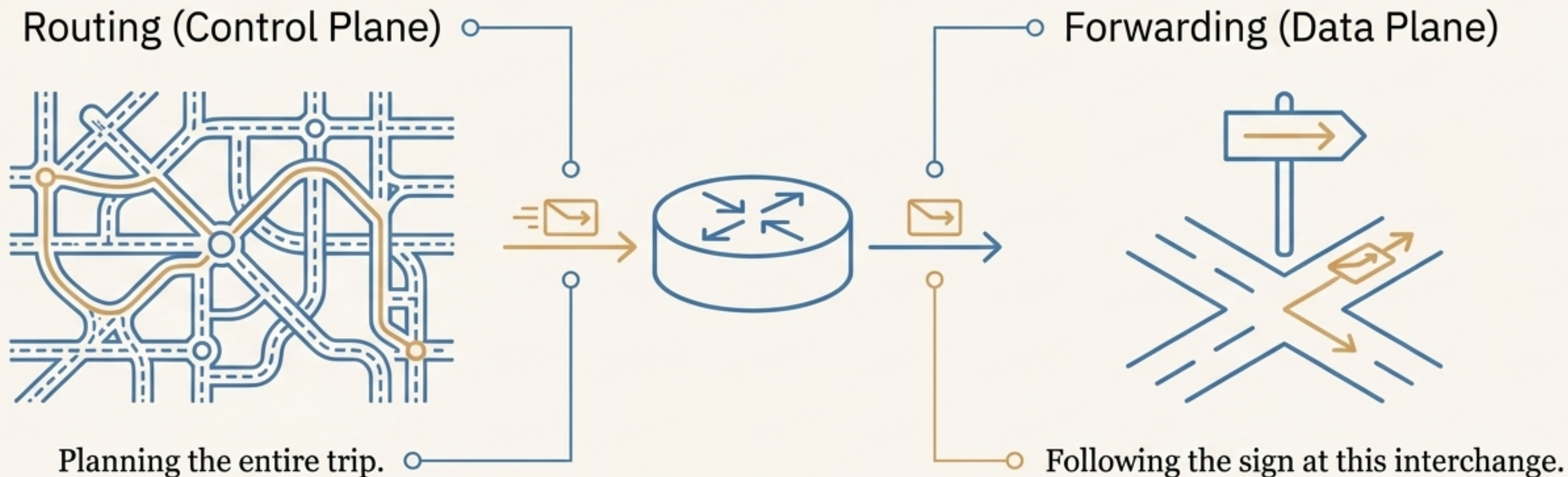


# Every Traveler Needs a Passport and an Itinerary





# The First Intersection: Making the Right Turn



The network-wide logic that determines the end-to-end path. Routing algorithms create the forwarding tables that forwarding uses.

The local, per-router function of moving a packet from an input link to the correct output link. This is our focus.



# Inside the Router: The Logic of the Forwarding Table

Arriving Packet Destination:

11001000 00010111  
00010110 10100001

Forwarding Table

| Destination Address Range (Prefix) | Link Interface |
|------------------------------------|----------------|
| 11001000 00010111 00010...         | 0              |
| 11001000 00010111 00011000...      | 1              |
| 11001000 00010111 00011...         | 2              |
| otherwise                          | 3              |

Match

Match

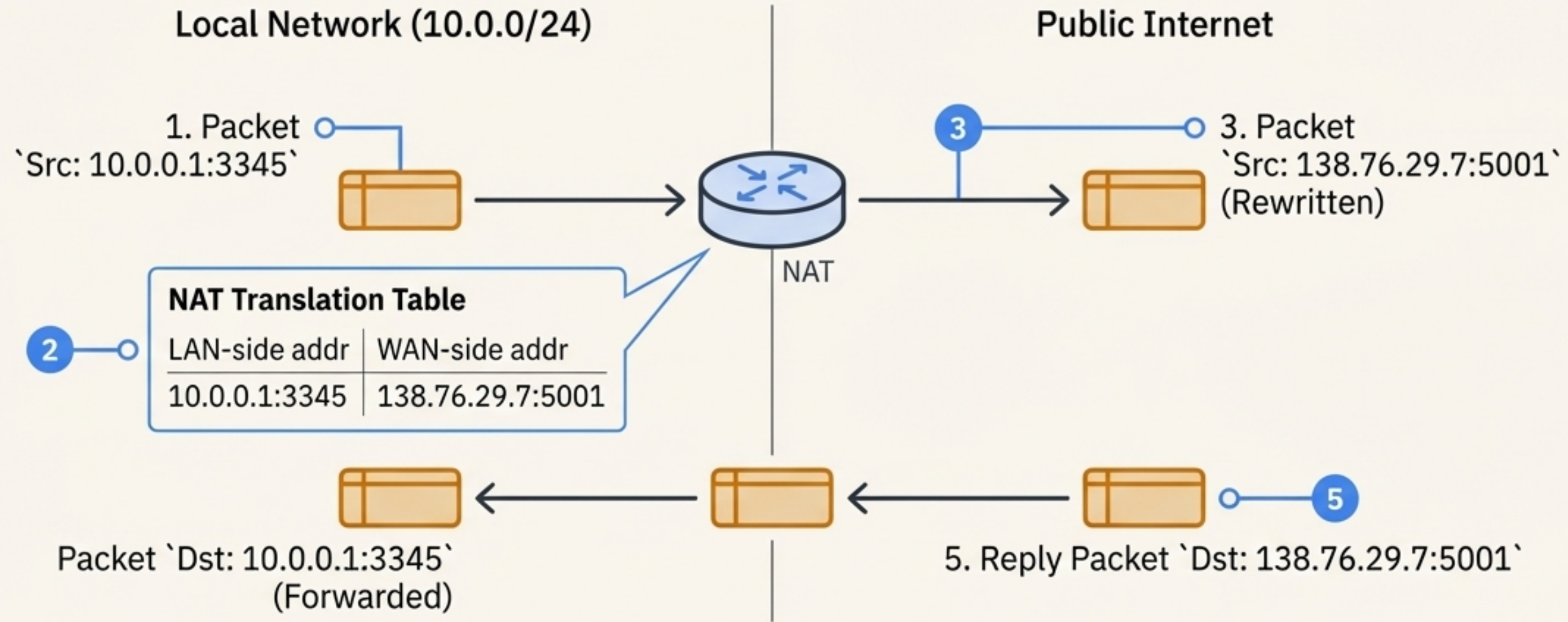
**Longest  
Prefix  
Match**

Packet  
forwarded to  
Interface 2

Routers use a forwarding table created by the control plane. To decide where to send a packet, they use the longest prefix matching rule: find the entry in the table that has the longest match with the packet's destination address. This is often accelerated using Ternary Content Addressable Memories (TCAMs).



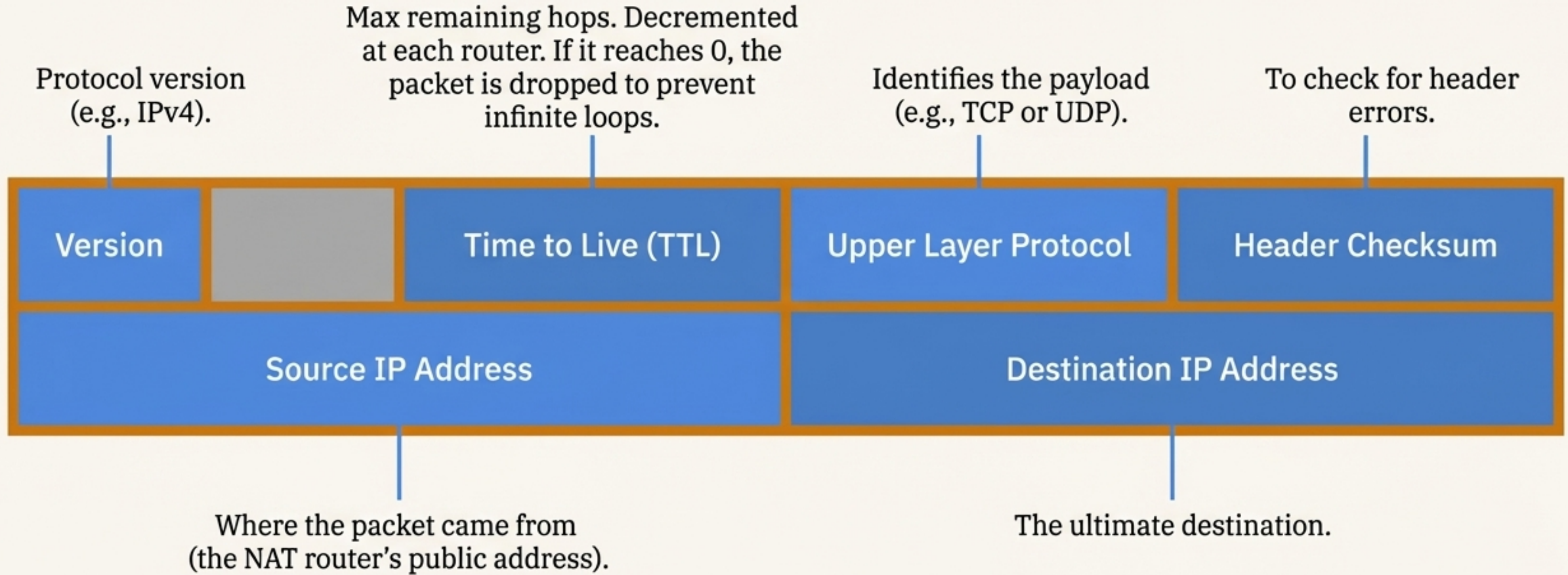
# Crossing the Border with Network Address Translation (NAT)



NAT allows all devices in a local network to share one public IPv4 address. The NAT router rewrites the source IP and port for outgoing datagrams and reverses the process for incoming replies. This was a crucial solution to the problem of IPv4 address exhaustion.



# The Anatomy of Our Traveler: The IP Datagram



The IP datagram header contains the critical information for the journey. Routers primarily examine the Destination Address to make forwarding decisions and decrement the Time to Live (TTL) at every hop.



# The Rules of the Road: The Internet's “Best-Effort” Service

Guaranteed  
Delivery?



Guaranteed  
Timing or Order?



Guaranteed  
Bandwidth?

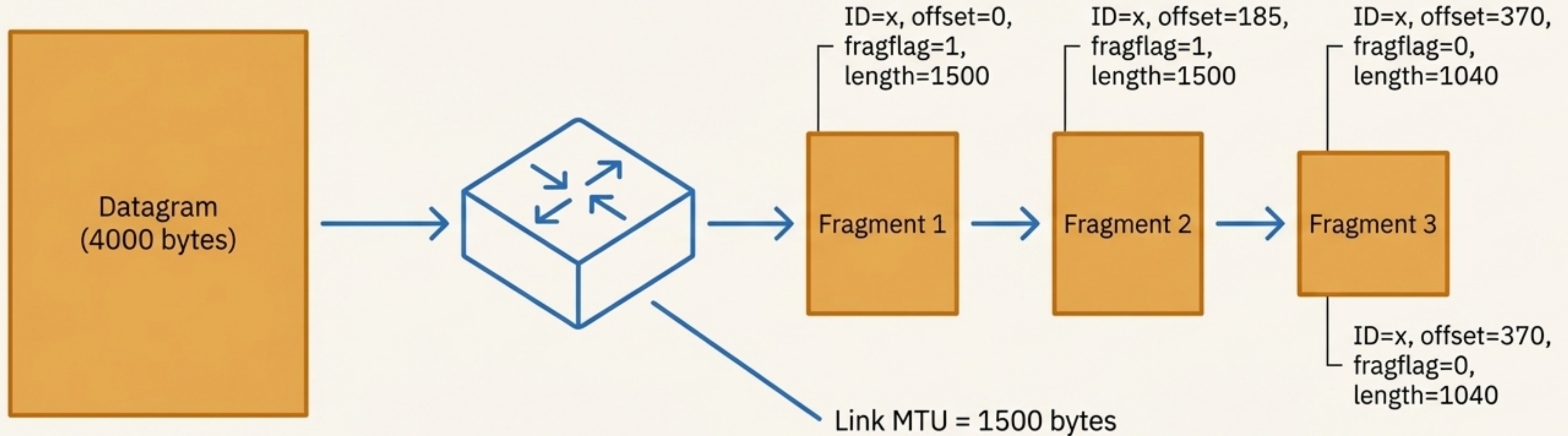


The Internet's network layer provides a “best-effort” service. It doesn't guarantee successful delivery, timing, or order. This simplicity is what allowed the internet to scale so successfully. For reliability, the network relies on transport-layer protocols like TCP.

**Insight:** Despite the lack of guarantees, sufficient bandwidth provisioning and application-layer services (like Content Distribution Networks) make performance “good enough” for most real-time applications, most of the time.



# A Necessary Detour: Fragmentation and Reassembly



Network links have a Maximum Transfer Unit (MTU). If a datagram is larger than the MTU of the next link, the router fragments it into several smaller datagrams. These fragments are only reassembled at the final destination host, using fields in the IP header (ID, offset, flags) to piece them back together.



# The Next Generation of Passports: The Rise of IPv6

## IPv4

32-bit address space (~4 billion addresses)

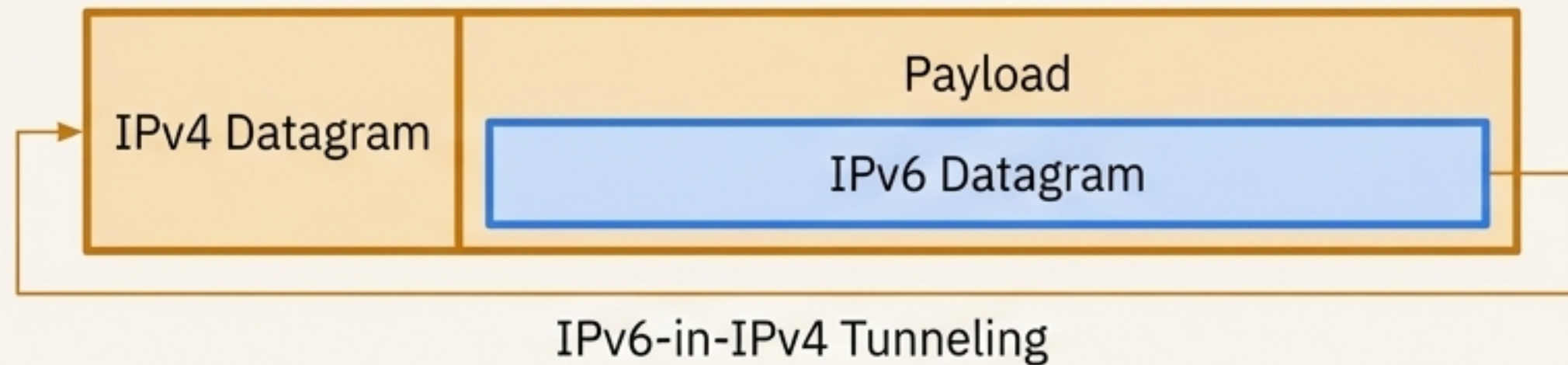
*“Who the hell knew how much address space we needed?”* - Vint Cerf

## IPv6

128-bit address space

- Fixed 40-byte header for faster processing
- No fragmentation allowed by routers
- No header checksum

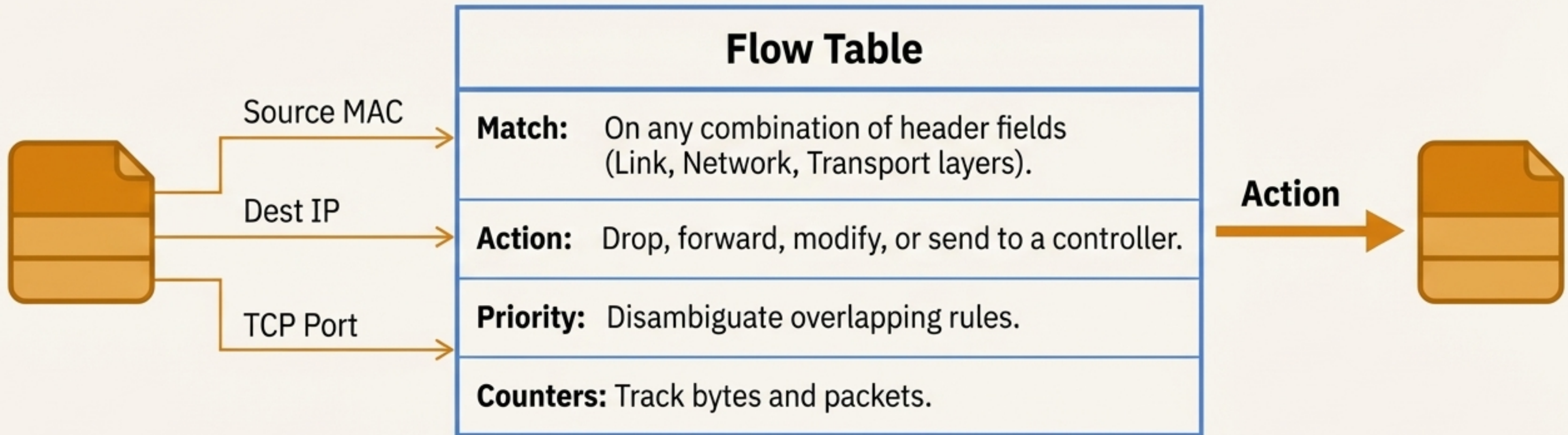
### Transitioning with Tunneling



**As of 2023, ~40% of Google's clients access their services via IPv6, but adoption has been a 25+ year process.**



# A Smarter Way to Travel: Generalized Forwarding

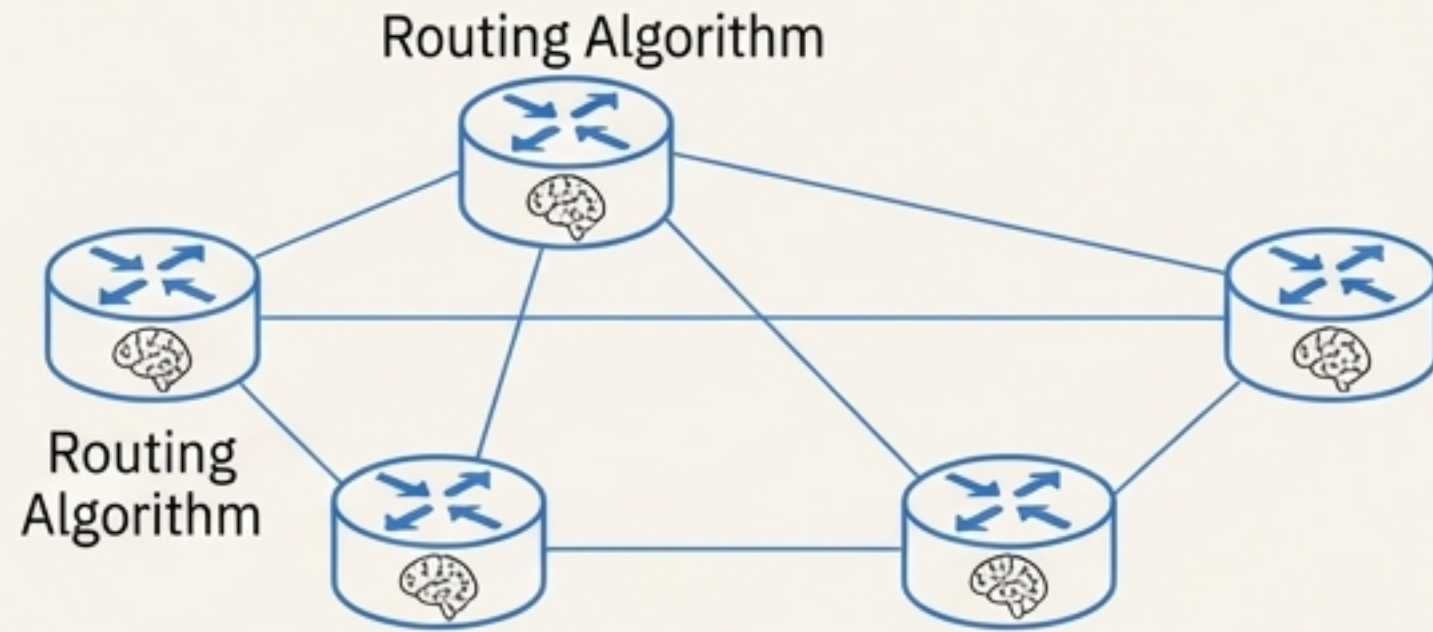


Traditional forwarding is based only on the destination IP address. Generalized forwarding abstracts this into a powerful **match-plus-action** model. Routers can make decisions based on numerous header fields, enabling functions like firewalls, NAT, and switches to be unified under a single abstraction.

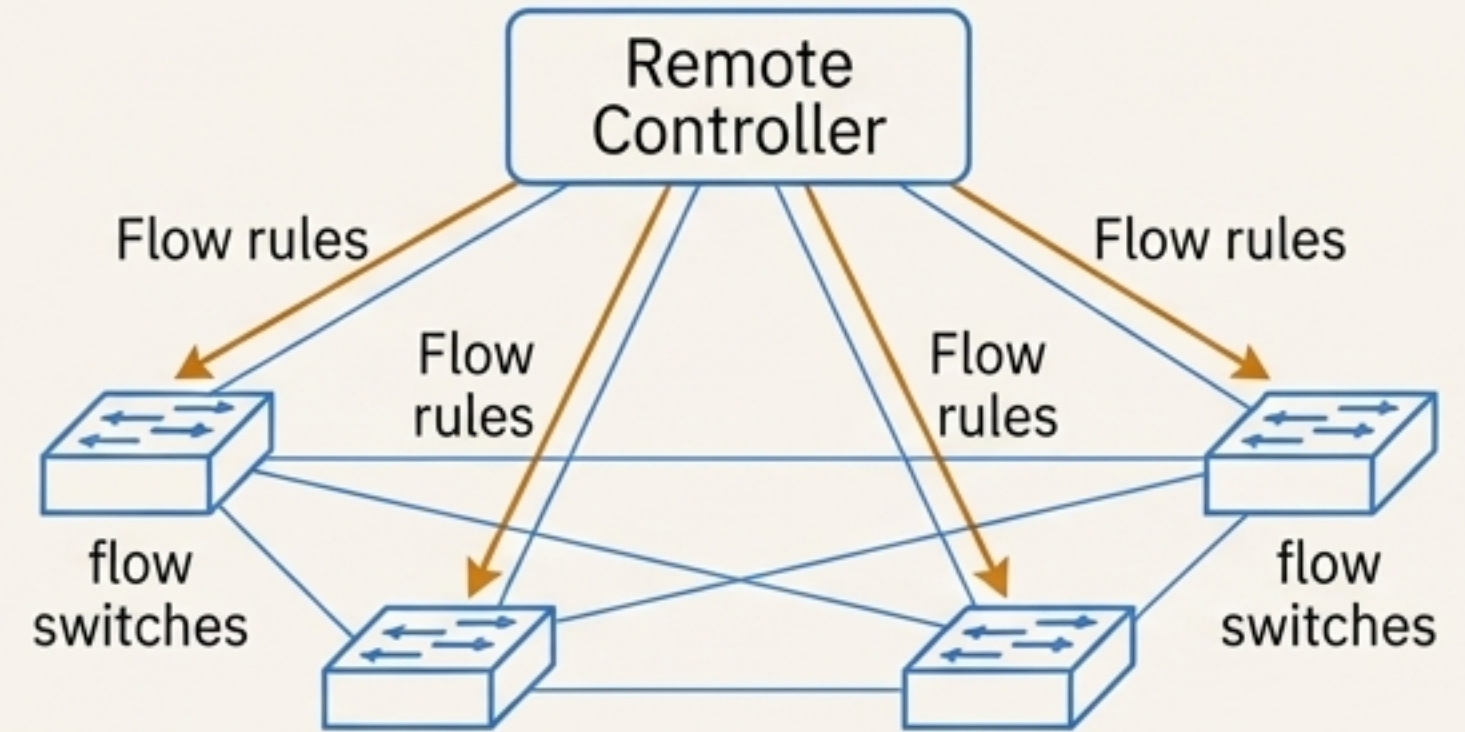


# From Road Signs to Central Traffic Control: Software-Defined Networking

Per-Router Control Plane



Software-Defined Networking (SDN) Control Plane



OpenFlow in Action

Match: IP Dst = 51.6.0.8

Action: forward(port 6)

With SDN and protocols like OpenFlow, the network's control plane is decoupled from the data plane. A central controller computes network-wide behavior and installs simple match-action rules in the forwarding devices. This provides unprecedented programmability and control over the packet's journey.